# SentiLingo

## Multilingual Customer Sentiment Analysis Platform

**Concept presentation**

DEPI Round 5 · Microsoft Machine Learning
Final presentation: 4 December 2026



# The customer-feedback problem

Useful signals are buried across languages and channels

## Feedback arrives everywhere

Reviews, surveys and support conversations create more text than a small team can read consistently.

## Arabic adds real variation

Formal Arabic and Egyptian expressions differ in spelling, vocabulary and tone.

## Manual tagging does not scale

Slow and inconsistent labels delay decisions and make quality difficult to measure.



# The SentiLingo concept

One reproducible path from multilingual messages to useful evidence



**English · MSA · Egyptian** | **Clean and standardise** | **Compare and select models** | **Label, confidence and evidence**

Concept only · measured performance will appear after reproducible experiments

# Supported scope

A narrow release that still tests multilingual and dialect-aware NLP

**3**

**language varieties**

- English
- Modern Standard Arabic
- Egyptian Arabic

**3**

**sentiment labels**

- Positive
- Neutral
- Negative

Deferred: Franco-Arabic, emotion detection and aspect-level sentiment

# Model development journey

Each stage must produce evidence before the next stage expands scope

01

### Prepare data

Licensed sources, common labels and frozen splits

02

### Classical baselines

Dummy, Naive Bayes, Logistic Regression and Linear SVM

03

### Advanced model

Fine-tune XLM-R Base; MARBERT remains controlled scope

04

### Select and lock

Validation chooses the complete configuration

05

### Evaluate once

Frozen test set, error analysis and local Streamlit demo

**Stop rule** Optional models and cloud deployment stop when they threaten evaluation, documentation or rehearsal.

# Fair evaluation and traceability

Strong English performance cannot hide weak Arabic performance

1

## Validation

Compare configurations using the mean of language-specific macro-F1 scores

2

## Lock

Freeze preprocessing, features, model and thresholds

3

## Test once

Report the frozen-test result and analyse errors

## Evidence recorded for every run

- Dataset and split versions
- Preprocessing and model configuration
- Seed, hardware and duration
- Metrics by language and sentiment
- Artefacts, checksums and Git commit

**No measured scores appear in this concept deck.**

# A multi-page local dashboard

Business workflows stay separate from model evidence

- **Overview**
- Analyze Text
- Analyze CSV
- Results
- Model Performance
- Experiment Registry
- Help and Limitations



Session-only predictions · Arabic right-to-left display · Offline presentation path

# Privacy boundary and data architecture

The registry preserves reproducibility without becoming a customer database

## Prediction session

### Processed in memory

Single messages and uploaded CSV rows remain available only for the active session and export.

**No customer prediction history**

## Experiment registry

### Reproducibility metadata

Dataset versions, experiment runs, metrics and artefact links create an auditable model history.

**No accounts, passwords or raw feedback**

Local-first delivery protects the demonstration from network, quota and cloud-service failures.

# Five-week delivery plan

The team protects a small complete release around full-time work schedules




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# What success looks like

A defensible learning project with evidence that recruiters and evaluators can inspect

### Reliable model evidence

Fair baselines, language-level metrics and honest error analysis

### Usable local product

Single-text and CSV workflows with Arabic support and clear limitations

### Reproducible portfolio

Versioned experiments, documentation, model card, video and presentation

## Project team


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**Current stage: planning and design complete · measured results will replace concept placeholders during implementation**